

RAG for Image Captioning

Saravanan S 24210090 IIT Gandhinagar 24210090@iitgn.ac.in	Maharshi Patel 24210059 IIT Gandhinagar maharshi.patel@iitgn.ac.in	Harsh Verma 24250039 IIT Gandhinagar harsh.verma@iitgn.ac.in
---	--	--

Abhiroop Chintalapudi 24250007 IIT Gandhinagar abhiroop.chintalapudi@iitgn.ac.in	Mithlesh Singla 24210063 IIT Gandhinagar mitlesh.singla@iitgn.ac.in
--	---

Uday Sankar Gottipalli 24210108 IIT Gandhinagar uday.gottipalli@iitgn.ac.in	Abhay Pisharodi 24290001 IIT Gandhinagar abhay.pisharodi@iitgn.ac.in
---	--

1 Report

This report follows the assignment specification (CS 613: Natural Language Processing) for project. The source code for our experiments will be available at [here](#)

1.1 Feedback and Novelties (10 Pts)

Since the first presentation, our team initially proposed to implement a multilingual image captioning RAG (Retrieval-Augmented Generation) model and make modifications to its architecture. However, based on the professor's feedback, we shifted our focus more towards improving the architecture of the RAG model itself instead of expanding it to multiple languages at this stage.

During our mentor discussions, it was suggested that we carefully analyze the baseline paper's implementation to identify the main bottlenecks and understand where improvements could be made. Through this process, we found that the baseline model had limitations in representing multiple objects within an image, which affected the quality of the generated captions and the retrieval results. We also noticed potential inefficiencies in the nearest embedding search algorithm and the way visual features were encoded.

To address these issues, we started working on enhancing the image representation by integrating object detection and segmentation algorithms. This allowed us to detect multiple objects in a single image and generate individual embeddings for each object using CLIP. We then combined these embeddings to form a more comprehensive image representation, which was later used for searching and retrieving relevant information from the database.

By doing this, we aimed to overcome the issue of limited object representation and improve the overall performance of the RAG model.

Throughout our weekly meetings, we continuously refined this approach, discussed various design choices, and compared the results of our modified architecture with the baseline outcomes.

Key novelties implemented:

- **Multi-object representation:** Integration of DETR-based object detection and semantic segmentation to identify and encode individual objects/regions within images
- **Enhanced retrieval context:** Generation of crop-specific embeddings that are individually queried against the ChromaDB database, providing richer contextual information
- **Vision-language generation:** Implementation of BLIP-2 for caption generation conditioned on both visual input and retrieved textual context
- **Ablation-ready architecture:** Three distinct approaches (original RAG, aggregation-based, and full vision-language) enabling systematic performance comparison

1.2 Baselines and State-of-the-Art (10 Pts)

We will compare against the following baselines and SOTA papers:

- DIR ("DIR: Retrieval-Augmented Image Captioning with Comprehensive Understanding" by Hao Wu et al.) — described in our template as the current state-of-the-art for robust retrieval-augmented image captioning. DIR

improves image representations using diffusion guidance; code was not available to us.

- SmallCap — available implementation; used as a practical baseline since DIR and other SOTA models build on SmallCap.
- RobustCap — we attempted to run the available code but were unable to get it working reliably; it remains a potential competitor if we resolve environment issues.

Baseline implementation availability: for this project, SmallCap code is available and used; DIR code is not available, so we re-implement comparable components where feasible.

1.3 Datasets (5 Pts)

Primary datasets and handling:

- COCO dataset (common choice for image captioning). Use standard train/val/test splits unless otherwise noted.
- Dataset Size: 243429 Captions
- Images: 40505
- Captions: 243429
- Avg Captions per image: 5
- Vocabulary size: 17,039 (as reported for COCO in related work).

1.4 Experimental Plan (15 Pts)

Our Experimental Plan

- Models/approaches:
 - Used ChromaDB for Retrieval-Augmented-Generation(RAG)
 - Used facebook/detr-resnet-50 and facebook/detr-resnet-50-panoptic for object detection and segmentation
 - Salesforce/blip2-opt-2.7b for Generating captions from images
- Each crop is sent to the database and captions are retrieved which are then sent to the LLM for captioning.

1.5 Project Management (10 Pts)

Team organisation and resources:

- Proposed novel solutions generation: iterative design based on baseline analysis, mentor feedback, and small-scale experiments.
- Compute resources: Used Colab and Singularity for Resources.
- Task distribution: Shared by Everyone.

2 Experiment

This section summarises the implementation and evaluation plan from the assignment prompt.

2.1 Implementation (50 Pts)

We implemented a complete RAG-based image captioning pipeline with the following key components:

Core Architecture:

- **Embedding Model:** CLIP (ViT-B-32) for image and region encoding
- **Retrieval Database:** ChromaDB populated with 10,000 COCO training image embeddings
- **Vision-Language Model:** BLIP-2 (Salesforce/blip2-opt-2.7b, 2.7B parameters) for caption generation conditioned on image and retrieved context

Advanced Image Understanding:

- **Object Detection:** DETR (facebook/detr-resnet-50) to extract salient object regions
- **Semantic Segmentation:** DETR Panoptic (facebook/detr-resnet-50-panoptic) for region-based crops
- **Multi-region Retrieval:** Each detected object/segment crop is individually encoded, captions retrieved, and aggregated with original image retrievals

Three Evaluated Approaches:

- **Original RAG:** Retrieval from original image only → BLIP-2 generation
- **Proposed RAG (Aggregation):** Text concatenation of original + crop-level retrievals (no LLM generation)

Method	Train Data	Flickr30k	NoCaps Val	Out-domain	Overall
Heavyweight training methods					
BLIP [17]	129M	446B	-	-	115.3
BLIP2 [18]	129M	1.2B	-	-	124.8
ViECap [12]	COCO	124M	47.9	13.6	66.2
Lightweight training methods					
MiniGPT4 [40]	5M	3.94M	78.4	16.9	110.8
ClipCap [24]	COCO	43M	-	-	49.1
SmallCap [29]	COCO	7M	60.6	-	-
EVCap [19]	COCO	3.97M	84.4	18.0	116.5
DIR (ours)	COCO	3.97M	85.7	18.2	120.7

Table 1: Summary of methods and their performance metrics.

- **Proposed RAG + BLIP-2:** Full vision-language generation using original + crop retrievals as contextual input

Technical Optimizations:

- Memory management: 8-bit quantization support, aggressive GPU cache clearing, reduced beam search (3 beams)
- GPU requirements: 6GB (without DETR), 16GB (with DETR enabled)
- Evaluation: 1000 COCO val2017 images, standard metrics (BLEU-1/2/3/4, METEOR, ROUGE-L, CIDEr) via pycocoevalcap

2.2 Reporting Results and Findings (10 Pts)

The following figure shows a comparison of the performance metrics between the two proposed RAG variants (Proposed RAG (Original + Crops) and RAG) across standard evaluation metrics.

2.2.1 Sample results

Below we provide two representative qualitative examples from our experiments. Each example shows the original retrieved captions (Original RAG), the crop-level captions and the final aggregated caption using the Proposed RAG (Original + Crops).

Example 1 – Image 000000261982.jpg Original retrieved captions (Original RAG):2

“A skateboarder is riding his skateboard at night. A young man riding a skateboard on a street. A skateboarder wearing a plaid shirt and jeans. A skateboarder at night preparing to skate away.”

Proposed RAG (Original + Crops):

Number of Crops Generated: 4

Crop Captions:

- **Crop 1:** A skateboarder is riding his skateboard at night. A young man riding a skateboard on a street. A skateboarder at night preparing to skate away.
- **Crop 2:** A tabby house cat walking all over it’s owner. Cat sitting on the stomach of a woman laying down on a bed. A cat looking down at the person lying on the bed.
- **Crop 3:** A skateboarder is riding his skateboard at night. A young man riding a skateboard on a street. A skateboarder wearing a plaid shirt and jeans.
- **Crop 4:** Small grey cat crawling over a woman’s stomach. Cat sitting on the stomach of a woman laying down on a bed. A cat looking down at the person lying on the bed.

Final Aggregated Caption: A skateboarder is riding his skateboard at night. A young man riding a skateboard on a street. A skateboarder wearing a plaid shirt and jeans. skateboarder preparing away. tabby house cat walking all over owner. sitting the stomach woman laying down bed. looking person.

Example 2 – Image 000000248400.jpg Original retrieved captions (Original RAG):3

“A man sits on a toilet in a bathroom. A man sits on a toilet while looking at porn

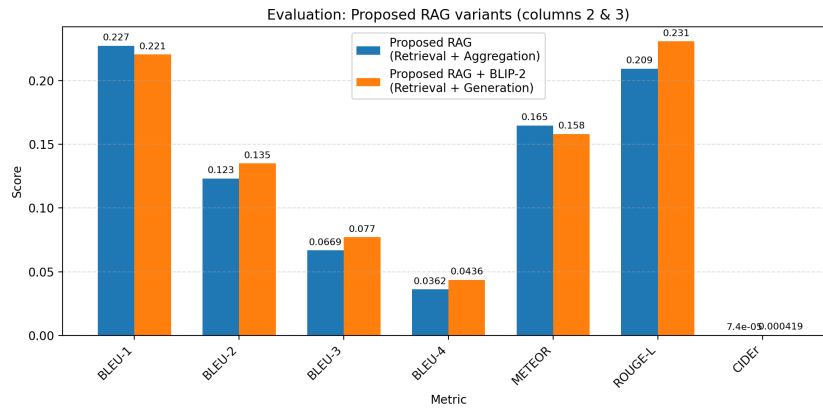


Figure 1: Performance comparison between Proposed RAG variants across evaluation metrics (BLEU, METEOR, ROUGE-L, CIDEr). The chart shows the quantitative results for both the Original+Crops approach and the Crops-only approach.



Figure 2: Example 1 (Image 000000261982) — original image and generated crops used by the Proposed RAG aggregation.

magazines. A person sits on the toilet in a bathroom reading a magazine. A fat ass sitting on a toilet with lady magazines.”

Proposed RAG (Original + Crops):

Number of Crops Generated: 4

Crop Captions:

- **Crop 1:** A room is shown with a washer and oven. An oven, and clothes washer are in a small kitchen. a kitchen with a stove a dih washer and a mcirowave.
- **Crop 2:** A pizza sitting on top of a table with olives, sausage and cheese. A pizza with multiple toppings on it, including olives. The personal pizza is topped with black olives and anchovies.
- **Crop 3:** A bathroom with urinals that have graffiti above them. A series of urinals on a graffiti covered

wall. Three urinals and another toilet in a bathroom.

- **Crop 4:** A dirty bathroom consisting of a toilet, bath tub and sink. a bathroom with a toilet and a sink. A bathroom mirror, with a person taking a picture in it.

Final Aggregated Caption: A man sits on a toilet in a bathroom. A man sits on a toilet while looking at porn magazines. A fat ass sitting on a toilet with lady magazines. room shown washer and oven. clothes are small kitchen. stove dih mcirowave pizza top table olives, sausage cheese. multiple toppings.

2.3 Viva (40 Pts)

Be prepared to demonstrate the code, run a small end-to-end example on a held-out image, and discuss the design choices, parameter counts, ablations, and failure cases.

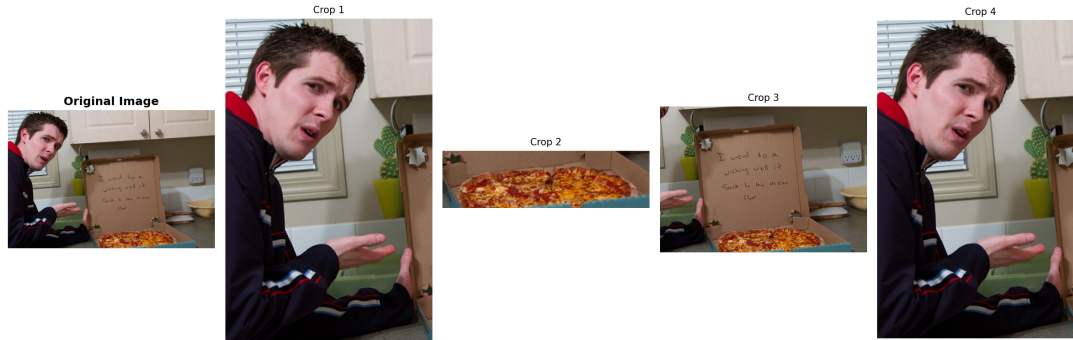


Figure 3: Example 2 (Image 000000248400) — original image and generated crops used by the Proposed RAG aggregation.

References

- Alfred V. Aho and Jeffrey D. Ullman. 1972. *The Theory of Parsing, Translation and Compiling*, volume 1. Prentice-Hall, Englewood Cliffs, NJ.
- American Psychological Association. 1983. *Publications Manual*. American Psychological Association, Washington, DC.
- Rie Kubota Ando and Tong Zhang. 2005. A framework for learning predictive structures from multiple tasks and unlabeled data. *Journal of Machine Learning Research*, 6:1817–1853.
- Galen Andrew and Jianfeng Gao. 2007. Scalable training of L1-regularized log-linear models. In *Proceedings of the 24th International Conference on Machine Learning*, pages 33–40.
- Ashok K. Chandra, Dexter C. Kozen, and Larry J. Stockmeyer. 1981. [Alternation](#). *Journal of the Association for Computing Machinery*, 28(1):114–133.
- Dan Gusfield. 1997. *Algorithms on Strings, Trees and Sequences*. Cambridge University Press, Cambridge, UK.
- Mohammad Sadegh Rasooli and Joel R. Tetreault. 2015. [Yara parser: A fast and accurate dependency parser](#). *Computing Research Repository*, arXiv:1503.06733. Version 2.
- Hao Wu, Zhihang Zhong, and Xiao Sun. 2024. Dir: Retrieval-augmented image captioning with comprehensive understanding. *arXiv preprint arXiv:2412.01115*.